Impact of Physical Imaging Parameters on Deep Learning Performance for Pneumonia Detection in Chest X-ray Images

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Abstract

This study investigates the effect of physical imaging parameters like radiation dose, detector blur, and spatial resolution on the performance of a machine learning model trained to identify pneumonia in chest X-ray images. Variations in these parameters may arise in clinical settings due to human or systemic error and may significantly influence image quality and subsequent diagnoses. To effectively evaluate the impact of these parameters, we simulate changes in imaging conditions with mathematical manipulations. We then analyze how these conditions affect the model's ability to classify pneumonia. By analyzing the model's robustness to degraded image quality, we hope to gain insight into reasoning behind the model's classification degradation and whether or not a sufficiently "good" model can be trained from "bad" images.

1 Introduction

Pneumonia is an infection of the lungs often caused by bacteria, fungi and viruses such as the flu, SARS-COV-2, or RSV [1]. Biologically, these infections lead to inflammation within the alveoli, filling them up with fluid or pus. This decreases the lung's ability to effectively perform gas exchange into the bloodstream. Symptoms of pneumonia include coughs, chest pain, fever, respiratory distress, among others [2].

Pneumonia is a leading cause of hospitalization in both adults and children. Annually in the US alone, over 40,000 deaths are due to pneumonia, of which the majority are adults over the age of 65 [1]. Globally, pneumonia is the leading cause of death for children under the age of 5, accounting for 16 percent of their deaths.

Currently, pneumonia is diagnosed through recording health history and physical exams. Physicians may also order additional tests like X-rays, pulse oximetry, blood tests, or sputum tests (citation). If chest X-ray imaging is ordered, the radiologist often looks for white spots in the lungs called infiltrates, which indicate the presence of fluid or other substances in the lungs, a key sign for the diagnosis of pneumonia [4].

In order to accurately model variations in radiation dosage, we set out to vary the Signal to Noise Ratio (SNR) of our chest X-ray images. The SNR of an X-ray image is directly proportional to the dosage, expressed by the relationship $SNR \propto \sqrt{Dose}$ [5]. By this relationship, as dosage decreases, the ratio of our signal to the noise received decreases as well, meaning a decrease in overall image quality. Since X-ray images are generated by effectively summing individual photons as they pass and scatter through the patient, a Poisson distribution can accurately model the random distribution of photons as they are received by the detector. While there are other sources of noise in X-ray imaging, such as from nuclear electronic noise or other external sources [6], this study focuses on the effect of noise from radiation dosage, which can be investigated with the Poisson distribution model.

In radiographic systems, detector blur arises from many different physical processes (i.e. scintillator light diffusion, electronic cross-talk, etc.). These effects result in broadening the system's response to a point-source which can be approximated by a Gaussian function. Different levels of

detector blur can be simulated by altering the standard deviation of the Gaussian kernel, enabling thorough analysis of the blur's impact on machine learning classification performance.. [7]

Detector resolution refers to the ability of an imaging system to resolve spatial details. This ability is largely determined by the detector pixel size. A detector with larger pixels results in lower spatial resolution, as these pixels integrate signals over a wider area. As a result, systems with larger detector pixel sizes tend to lose high spatial frequency content. In chest X-Ray images, pneumonia often appears as small, localized opacities. These appear in images as subtle intensity changes over short spatial distances. Essentially, pneumonia is detectable through high spatial frequency content. This is why it is crucial to determine the effects that various detector pixel sizes have on the ability of a Convolutional Neural Network (CNN) to detect pneumonia.

While CNNs are very robust, given a large enough training set and well-tuned hyperparameters, these models can identify the wrong patterns, if the hardware used upstream does not capture the important details related to the disease.

2 Related work

With the rapid development of the field of Artificial Intelligence (AI) and Machine Learning (ML) models, there have been many attempts at using these models for diagnosis, prediction, and treatment plans [8]. One such paper [4], approaches a very similar problem as is examined in this study. Their study, however, employs deep transfer learning and designed three different CNN models, as opposed to our one. As an expansion of this study, looking further into the performance of different types and combinations of models could generate more robust results in the future.

3 Methodology

We use a publicly available pediatric chest X-ray dataset from Guangzhou Women and Children's Medical Center, originally introduced by Daniel S. Kermany et al., containing 5,683 images labeled as normal or pneumonia with multi-expert validation. A DenseNet-121 CNN was trained on the original images to generate baseline metrics. To address the stark class imbalance in the dataset, a class-weighted cross-entropy loss was used, with weights proportional to the inverse class frequencies.

Following the initial run and generation of the baseline weights, our goal was to test this base model against images modified according to three common physical imaging parameters that affect X-ray images: radiation dosage, detector blur, and the spatial resolution of the detector.

Each of the degradation techniques used in these experiments apply different types of degradation to the original images. As a result, it is difficult to compare the extent of each modification against each other. To account for this, Structural Similarity Index (SSIM) was implemented on all of the degraded images as a proxy for equivalent amounts of degradation. For mild, medium and heavy datasets across noise, we established baseline SSIM values of 0.9, 0.8, and 0.7, respectively. We then ran a loop on 20 images that adjusted all of the control parameters for each type of degradation, until the SSIM value for each degradation fell within .05 of the target baseline value.

Utilizing the generated SSIM value for the radiation dosage parameter, a Poisson noise distribution was applied as a weight for each pixel of the original image. We repeated this process with three different scaling factors, 0.4914, 0.5856, 0.7151, to simulate a low, medium, and high dosage, respectively.

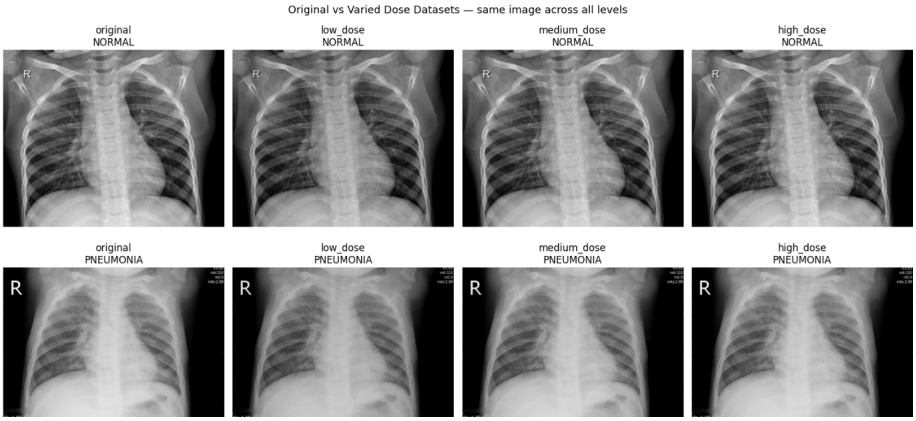

Figure 1: Modified Images with Varying Radiation Dosage

The simulation of detector blur in X-ray imaging was accomplished by convolving high-resolution X-ray images with a Gaussian kernel or Point Spread Function (PSF). By altering the standard deviation values of the Gaussian kernel, we were able to simulate varying levels of detector blur [7]. This enables us to analyze the impact of different strengths of detector blur on the performance of the machine learning classification model. Specifically, to modify images in accordance with the SSIM, we introduced standard deviation values of 1.6950, 2.8529, and 4.4025 for mild, medium, and heavy alterations, respectively.

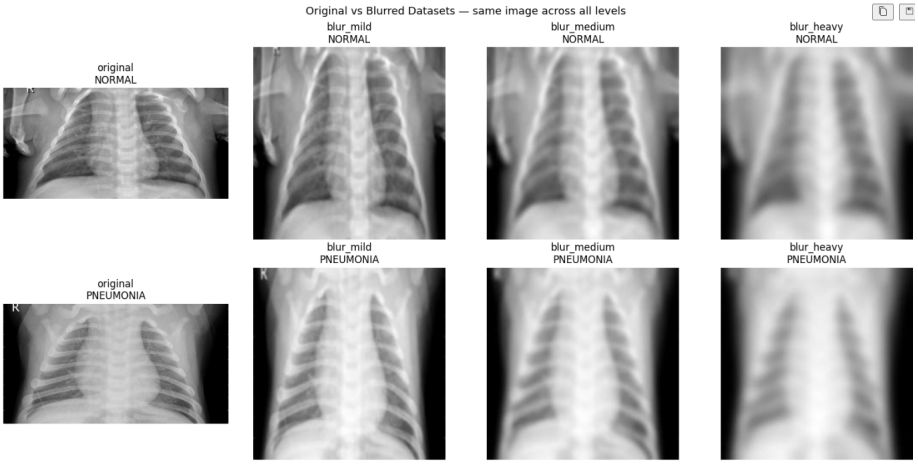

Figure 2: Modified Images with Varying Detector Blur

A downsampling-upsampling pipeline was implemented to simulate the effects of varying detector pixel size. Images were downsampled by a scale factor (determined via SSIM matching) to reduce spatial resolution and attenuate high-frequency information, then upsampled to resize the image back to its original dimensions (224x224). Bilinear interpolation was used for resizing due to its smoothing (low-pass) properties and similarity to detector integration.

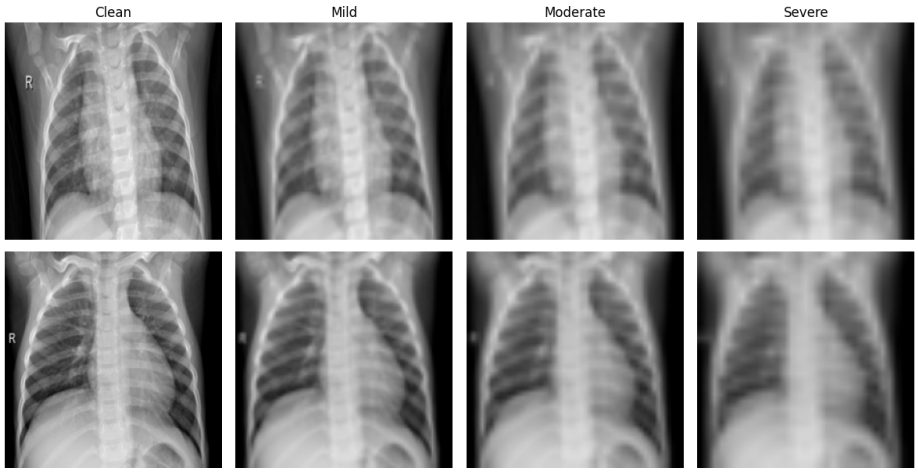

Figure 3: Modified Images with Varying Detector Crystal Size

Following each image degradation process, modified image class labels were predicted using the baseline model to assess its ability to generalize. After determining these results, we retrained our model on each individual set of modified images and retested with the associated data sets.

4 Results

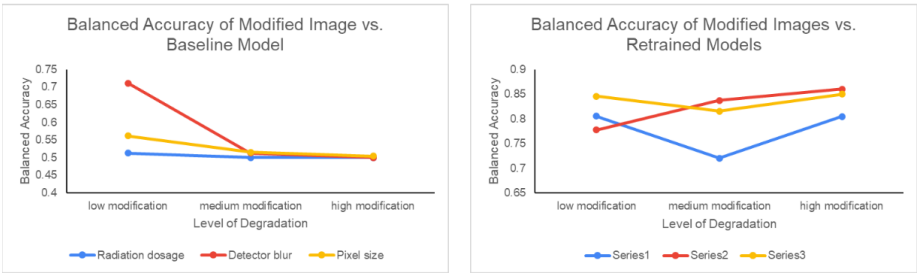

Figure 4: Balanced Accuracy Results

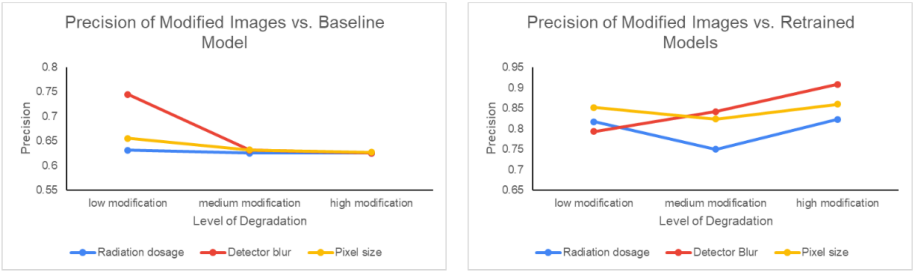

Figure 5: Precision Results

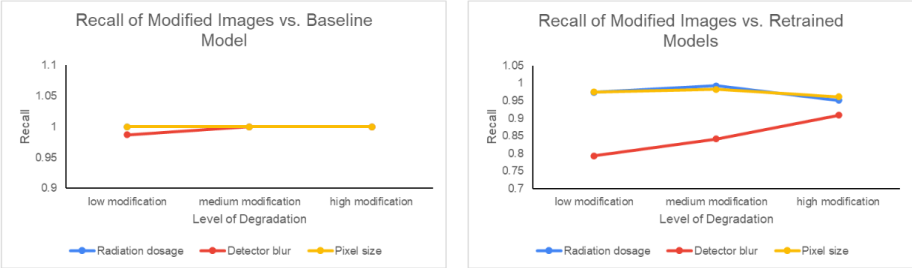

Figure 6: Recall Results

When the radiation dosage modified images were tested against the baseline weights, the performance decreased starkly, with an AUC of .67 and .73 for medium and low dosage, respectively. However, the AUC of the high dosage image remained at a strong 0.93. Though, all three radiation dosage accuracies dropped to around 0.63 with a balanced accuracy of around 0.5. After retraining with the individual varied dosage datasets, these statistics improved drastically. The AUC for the dosages were .94, .91, and .91 for the high, medium, and low dosages respectively. The balanced accuracies increased to .81, .72, and .80, which is an increase but not to the same performance of the untouched images.

Testing the baseline weights with Gaussian blur-modified images demonstrated a decrease in AUC; with unmodified images resulting in an AUC of .8780 and decreasing to .8381, .7993, and .8005 for mid, medium and heavy levels of Gaussian blur, respectively. All three accuracy values did drop dramatically, however, from a clean accuracy value of .7644 to .6619, .6314, and .6250 for mild, medium, and heavy modifications. When training and testing with varied detector blur datasets, the AUC, as well as accuracy, remained rather consistent, hovering around .93 and .83, respectively. This is as would be expected when training and testing on the exact same type of dataset without any further modification as the model would've been trained on very specific labels and would be able to adjust the weights depending on each blurred image rather than have to guess based on the clear images.

The model's performance remained strong across all levels of pixel size related degradation, with test AUCs around .95 to .96. This indicates robust discrimination of the model even under resolution loss. Accuracy stayed relatively stable across all degradation levels with values .878, .857, and .878 for mild, moderate, and severe degradation, respectively. However, balanced accuracy is consistently lower than overall accuracy (.81-.85), indicating class imbalance still impacts performance despite a weighted loss function.

5 Discussion

SSIM was used as a proxy for relative degradation of images across types of degradation. SSIM is a metric used to measure the perceived similarity between two images, rather than just the pixel error. It evaluates luminance, contrast, and structure and outputs a value between 0 and 1, with values closer to 1 indicating that the images have closer perceived similarity. Although SSIM does not guarantee that degradation amounts are equivalent across distortion types, it serves as a useful standard to ensure consistent relative changes in perceptual similarity. This allows our study to focus on the impact of the different modes of degradation, rather than their individual amounts.

For all three of our selected physical simulations, we saw a drastic decrease in performance. This makes sense when considering what the model is trained to "look for." If it is identifying blurry, bright regions or areas with a decreased SNR, it will classify it as having pneumonia. With each of our modifications, we were able to visibly see an increase in noise or blurring across our images, whether it be in a Gaussian shape or due to the downsampling effect. It makes sense that our system will classify pneumonia much more frequently with these modified images, which is evident based on our results. For the detector blur dataset, we have a recall average of around 1 and a precision

ranging from .625 to .74. Similarly, for both radiation dosage and downsampling datasets, we see a recall of 1 and a precision around .625, which correlates to the ratio of pneumonia to normal images we have in our dataset. This means our original weights classified every image as pneumonia when they were degraded. Because these physical modifications are common in radiographic imaging, it is important that their variation doesn't drastically increase the false positive rate of detection. While a degradation in performance is not ideal, it is preferred over an increase in false negative rate of detection as a missed diagnosis can lead to delayed responses in treatment.

After retraining with the modified images, we see a stark jump in performance by the model with our test datasets. As the model "relearns" how to identify pneumonia with added blurring, it's able to more consistently identify when pneumonia is not present, rather than classifying as positive every time.

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